ELEC 872 - Replication Lab 2: DeepConvLSTM

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1 Discussion

1.1 Part 0: Reference model

I trained the reference model with default settings. It reaches weighted F1 0.8806 and macro F1 0.5368 on the test set. The large gap between the two is the class-imbalance. The *Null* class scores F1 0.943 while several drawer gestures stay low (*Close Drawer 2* 0.160, *Open Drawer 1* 0.252). This reference is the baseline for every part below.

1.2 Part A: Is the LSTM what helps?

I replaced the LSTM with two alternatives: `CNN` (CNN instead of recurrence) and `GRU` (GRU cells instead of LSTM).

Table 1: Part A results

Variant	Weighted F1	Macro F1
ref (LSTM)	0.8806	0.5368
`CNN`	0.8691	0.4496
`GRU`	0.8930	0.6175

Recurrence helps. The CNN has the lowest macro F1 (0.4496 vs. 0.5368) and fails the hardest gestures: it scores F1 $= 0.000$ on *Close Drawer 2* and 0.157 on *Close Drawer 1*. This confirms the paper's claim that modelling temporal dynamics improves recognition. However, the benefit is from recurrence in general, not the LSTM specifically: the GRU beats the LSTM on both metrics (0.6175 vs. 0.5368 macro). Weighted F1 barely moves for the CNN because it is dominated by *Null* (83% of windows, CNN *Null* F1 0.949).

1.3 Part B: Do the convolutions matter?

I run the experiments on the number of convolutional layers from four to two and one.

Weighted F1 is essentially flat (~ 0.88) across one, two and four conv layers, so removing convolutions did not hurt overall recognition on this run.

Table 2: Part B results

Variant	Weighted F1	Macro F1
ref (4 conv)	0.8806	0.5368
`Conv2`	0.8881	0.6086
`Conv1`	0.8875	0.5891

The macro F1 of the four-layer reference (0.5368) is actually *lower* than the one and two-layer variants. The honest conclusion is that within one and four layers, conv depth had little effect here.

1.4 Part C: How long should the window be?

I ran a short window (`window 20`) and a long window (`window 60`).

Table 3: Part C results

Variant	Weighted F1	Macro F1
ref (0.8s)	0.8806	0.5368
`w20`	0.8980	0.6152
`w60`	0.8944	0.5949

Both windows improve on the reference, and the short window is best overall. The paper highlights drawer gestures specifically:

Table 4: Per-gesture F1 for drawer gestures

Gesture	ref (0.8 s)	`w20`	`w60`
Open Drawer 1	0.252	0.337	0.387
Close Drawer 1	0.370	0.422	0.420
Open Drawer 2	0.325	0.329	0.314
Close Drawer 2	0.160	0.478	0.375
Open Drawer 3	0.550	0.612	0.594
Close Drawer 3	0.436	0.609	0.574

The short window gives the largest gains (*Close Drawer 2* $0.160 \rightarrow 0.478$, *Close Drawer 3* $0.436 \rightarrow 0.609$). I think this matches the paper's claim that

short drawer gestures do better when the window fits them; the 2 s window does not help drawers more than the 0.67 s one.

1.5 Part D: How much regularization is needed?

The paper code uses dropout 0.5 and nothing else, so in this code weight decay is already 0 and augmentation is already off by default. I therefore ran: remove dropout, remove weight decay, and add augmentation.

Table 5: Part D results.

Variant	Weighted F1	Macro F1
ref (dropout 0.5)	0.8806	0.5368
`no_dropout`	0.8911	0.5846
`no_weightdecay`	0.8886	0.6057
`with_augment`	0.8852	0.5851

Only the dropout run is a genuine change from the paper's setup; `no_weightdecay` is effectively the reference configuration (weight decay was never on), so its difference is seed noise. Removing dropout did not hurt (weighted 0.8911, macro 0.5846, both slightly above the reference), and adding augmentation did not help. I think this supports the paper's claim that regularization settings matter less than the architecture.

1.6 Part E: Does warm-up make training more stable?

I added a linear learning-rate warm-up over the first 5 epochs (`warmup-epochs 5`) and compared against the reference.

Table 6: Part E results

Variant	Weighted F1	Macro F1
ref (no warm-up)	0.8806	0.5368
`E_warmup`	0.8925	0.6138

Warm-up makes early training more stable. In the validation loss spikes sharply near epoch 27; the early epochs descend smoothly without that oscillation, and validation macro F1 climbs more steadily. Final macro F1 is also higher (0.6138 vs. 0.5368). The one caveat is that the warm-up run's validation loss drifts upward later in training, but the early-training stability the question asks about is clearly improved.

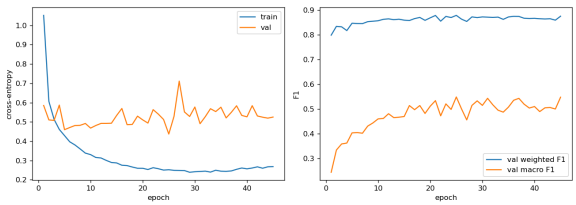


Figure 1: Reference learning curves.

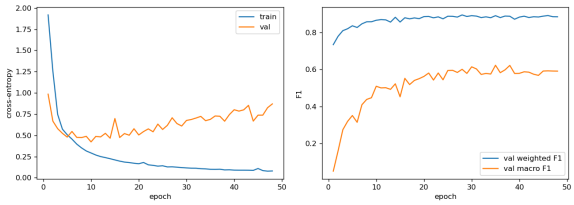


Figure 2: Warm-up learning curves.

1.7 Part F: Does focal loss help rare gestures?

I replaced cross-entropy with Focal Loss ($\gamma = 2$, `loss focal`).

Table 7: Part F results.

Variant	Weighted F1	Macro F1
ref (cross-entropy)	0.8806	0.5368
`Focal`	0.8873	0.5886

Focal loss raises macro F1 ($0.5368 \rightarrow 0.5886$) while leaving weighted F1 essentially unchanged ($0.8806 \rightarrow 0.8873$). The gain comes from the rare gestures, exactly as intended.

Focal loss down-weights the easy, dominant *Null* class and focuses learning on hard, rare gestures, nearly doubling F1 on *Open Drawer 1* and *Close Drawer 2*. Compared to the paper, which ignores the imbalance and reports weighted F1, focal loss keeps weighted F1 comparable while improving the macro F1 that reflects rare-class performance.

1.8 Conclusion

Across the six ablations, the design choices that actually mattered were: recurrence (Part A: removing it drops macro F1 and kills drawer gestures, though GRU works as well as LSTM); a window short enough to fit short gestures (Part C); and the loss function when rare classes matter (Part F: focal loss lifts macro F1 at no cost to weighted F1). Conv depth (Part B) and regularization settings (Part D) had little effect here, consistent with the paper's view that architecture matters more than these knobs, and warm-up (Part E) improved

Table 8: Per-gesture F1, cross-entropy vs. focal loss.

Gesture	ref	F_focal
Open Drawer 1	0.252	0.437
Close Drawer 1	0.370	0.429
Close Drawer 2	0.160	0.359
Open Drawer 3	0.550	0.614
Close Drawer 3	0.436	0.581

early-training stability. Throughout, weighted F1 stayed near 0.88 because the *Null* class dominates it; macro F1 was the metric that revealed which changes actually helped the rare gestures.

The code files are available in the DeepConvLSTM repository: https://github.com/yeasintaha/DeepConvLSTM.git